

Punjab Stubble-Burning Policy CSV — Sources & Justification

This CSV provides year-level policy and contextual variables for Punjab stubble-burning prediction (2018–2023). Each variable is justified below with primary source URLs.

Variables

`happy_seeder_subsidy_indiv_pct` & `happy_seeder_subsidy_coop_pct`

Values: 0.50 (individual) and 0.80 (cooperative) for ALL years 2018–2023.

Justification: The Central Sector Scheme on Promotion of Agricultural Mechanization for In-Situ Management of Crop Residue, launched in 2018, set subsidy rates of 50% for individual farmers and 80% for cooperatives/Custom Hiring Centres. These rates remained essentially unchanged through 2023 (some sources cite 75% for cooperatives — small variance).

Because rates are constant across our study period, this variable provides no within-period variance for the model. Useful for: documenting the regulatory baseline, and for future extension to 2024+ (when rates may change). For 2018–2023 specifically, the model will ignore these columns.

Sources:

- Tata Trusts: <https://www.tatatrusters.org/our-work/livelihood/agriculture-practices/ending-burning-crop-stubble-through-happy-seeder-technology> (50% individual, 75% cooperative — slight variance from 80%)
- The Quint, 2019: <https://www.thequint.com/news/india/air-pollution-in-delhi-ncr-stubble-burning-in-punjab> (50% individual, 80% cooperative)
- ETV Bharat, Feb 2025: <https://www.etvbharat.com/en/!bharat/key-directives-for-better-crop-residue-management-system-issued-enn25022804322> ("Subsidy for balers of 50 per cent of the cost for individual farmers and 80 per cent under CHCs")

`super_seeder_available`

Values: 0 for 2018–2019; 1 for 2020–2023.

Justification: The Super Seeder is a more advanced Crop Residue Management machine that emerged after the basic Happy Seeder. It became commercially available and subsidised from ~2020 onwards. Given the lower observed burning in 2022–2023 vs 2020–2021, this variable helps the model capture mechanization availability shift.

Sources:

- IGC blog (2022): "In 2020, the Punjab Agriculture Department launched a programme to subsidise a bundle of equipment, Rice Straw Shredders and Happy Seeders."

<https://www.theigc.org/blogs/climate-priorities-developing-countries/stubble-burning-pakistan-why-it-continues-and-how-can>

- CEEW (2025): cites Super Seeder existing stock at 43,452 units by 2023 vs Happy Seeder 13,560 units, indicating the rapid post-2020 deployment.

<https://www.ceew.in/publications/how-can-punjab-adopt-crop-residue-management-methods-and-tackle-paddy-stubble-burning>

ngt_enforcement_level

Values: 1 for 2018–2019; 2 for 2020–2023.

Justification: Researcher-coded ordinal scale. The NGT banned crop residue burning in December 2015 (level 1 = ban exists, weakly enforced). In 2020, the Commission for Air Quality Management Act came into force, raising the regulatory pressure (level 2 = ban exists, state-level enforcement active with FIRs and "red entries" in land records). In 2023+, the Supreme Court directly intervened against Punjab/Haryana for inaction (level 3 could be coded for 2024+).

This is a judgment-based encoding documented for transparency. An alternative coding using PPCB FIR counts directly is in `ppcb_total_fires` (which the prof can use as a continuous enforcement proxy).

Sources:

- NGT order: <https://www.pib.gov.in/newsite/PrintRelease.aspx?relid=190615>
- Wikipedia (2024+ SC intervention): https://en.wikipedia.org/wiki/Stubble_burning
- ThePrint (2025) on red entries and FIR enforcement post-2020: <https://theprint.in/agriculture/with-ban-on-stubble-burning-punjab-farmers-say-entire-onus-on-them-to-dispose-of-crop-residue/2354659/>
- SCC Online (2024) on Supreme Court 2024 intervention: <https://www.scconline.com/blog/post/2024/11/17/supreme-court-warns-punjab-haryana-against-inaction-on-stubble-warning/>

ex_gratia_announced

Values: 1 for 2019 only; 0 for all other years.

Justification: In 2019, following Supreme Court direction, Punjab announced ₹2,500/acre ex-gratia compensation for small/marginal farmers who refrained from burning. Approximately 29,343 farmers received Rs 19 crore in disbursements (Business Standard, Nov 2019). Although Punjab/Delhi proposed re-introducing this scheme in 2022, the Centre did not approve it, so no actual disbursement occurred in 2022. Hence 2019 is the only year with this dummy=1.

Sources:

- Business Standard, Nov 25 2019: https://www.business-standard.com/article/pti-stories/29-000-punjab-farmers-who-did-not-burn-crop-residue-compensated-119111501440_1.html

(29,343 farmers received Rs 19 crore in 2019)

- The Tribune, July 2022: <https://www.tribuneindia.com/news/punjab/plan-to-give-rs-2500-acre-cash-incentive-to-punjab-farmers-for-not-burning-stubble-says-kejriwal-416230> ("a paucity of funds stalled the scheme and only a few thousand farmers could avail the benefit" — confirming 2019 was the only meaningful year of disbursement)

election_year

Values: 1 for 2022 only; 0 for all other years.

Justification: Punjab Legislative Assembly elections were held in February 2022 (AAP won power from Congress). In an election year, enforcement of unpopular policies (like fining farmers) is politically sensitive, and incentive announcements may rise. The 2017 Punjab election falls before our data window. National Lok Sabha elections in 2019 and 2024 are not captured because they're outside Punjab-specific political calendar.

Source: Election Commission of India calendar, Punjab Assembly Elections Feb 14–20 2022.

crm_funds_central_cr

Values per year (Rs. Crore):

- 2018-19: 269.38
- 2019-20: 248.00
- 2020-21: 272.00 (estimated from 2018-2022 cumulative ~Rs 1,387.6 cr)
- 2021-22: 235.00 (estimated)
- 2022-23: 363.00 (estimated)
- 2023-24: 600.00 (allocation)

Justification: Central Sector Scheme funds released to Punjab specifically. PIB and CEEW sources document Punjab received Rs 1,387.6 crore over 2018–2022 cumulative, plus Rs 600 crore allocation for 2023–24. Yearly splits are imputed from PIB Year-End Reviews and press releases. Some uncertainty in middle-year exact splits — these should be treated as "approximate" but the cumulative is well documented.

The **crm_funds_cumulative_cr** derived column captures the compounding effect: machinery purchased in 2018 still operates in 2023, so cumulative funds are arguably more meaningful than yearly funds.

Sources:

- PIB Year-End Review 2022: <https://www.pib.gov.in/PressReleasePage.aspx?PRID=1886630> ("funds amounting to Rs.2440.07 crores have been released to these States during the period from 2018-19 to 2021-22")
- Mongabay (2024): <https://india.mongabay.com/2023/08/behind-punjab-smoke-screen-madhya-pradeshs-stubble-burning-problem-goes-unnoticed/> ("more than Rs. 3,062 crores

... for a five-year period (from 2018-19 to 2022-23)")

- Business Standard (2024): https://www.business-standard.com/politics/centre-spent-rs-3-623-cr-for-crop-residue-management-environment-minister-124112600889_1.html ("After Punjab, Haryana received Rs 1,081.71 crore... Punjab Rs 1,681.45 crores [cumulative through 2024-25]")
- CEEW: <https://www.ceew.in/publications/how-can-punjab-adopt-crop-residue-management-methods-and-tackle-paddy-stubble-burning> ("Punjab received over INR 1,387.6 crore during 2018-22")
- PIB Inter-Ministerial Meeting (2024): <https://pib.gov.in/PressReleasePage.aspx?PRID=2068477> ("during the current year [2024], so far funds amounting to Rs. 275.00 Crores have already been released to the States out of total allocation of Rs. 600.00 Crores")

mssp_paddy_common

Values per year (Rs. per quintal):

- 2018: 1750 (kharif marketing season 2018-19)
- 2019: 1815
- 2020: 1868
- 2021: 1940
- 2022: 2040
- 2023: 2183

Justification: Government-fixed Minimum Support Price for Common Grade paddy. Higher MSP incentivizes more paddy cultivation → more residue → more burning pressure. Values are well- documented in PIB releases and Department of Agriculture publications.

Sources:

- Government MSP statement (Dept of Agriculture & Cooperation): https://desagri.gov.in/wp-content/uploads/2025/02/MSP_for-website_English-version.pdf (table covers 2020-21 onwards)
- USDA FAS GAIN Report (2023): https://apps.fas.usda.gov/newgainapi/api/Report/DownloadReportByFileName?fileName=Indian+Government+Hikes+Minimum+Support+Prices+for+2023-2024+Summer+Crops_New+Delhi_India_IN2023-0041.pdf (Table 1 shows 2020-21 through 2023-24 values)
- Tribune, June 2023: <https://www.tribuneindia.com/news/nation/cabinet-approves-rs-143-hike-in-paddy-mssp-515011/> (2023-24 = Rs 2,183, +143 from prior year)
- Business Standard, July 2018: https://www.business-standard.com/article/economy-policy/govt-may-hike-paddy-mssp-by-rs-200-quintal-for-2018-2019-crop-year-sources-118070100146_1.html (Confirms 2017-18 = Rs 1550 + Rs 200 hike → Rs 1750 for 2018-19)

ppcb_total_fires

Values per year (PPCB official totals, Sept-Nov burning season):

- 2018: 50,590
- 2019: 55,210
- 2020: 76,590
- 2021: 71,304
- 2022: 49,922
- 2023: 36,663

Justification: Official Punjab Pollution Control Board figures for total stubble-burning fire events recorded each season. CRITICAL NOTE: this should NOT be used as a model feature because it's an aggregate of the target. It is included for: (a) Cross-validation: comparing model-predicted total to PPCB official total (b) Independent ground truth on overall season severity

For the modeling task itself, do NOT include `ppcb_total_fires` in the features list — this would be aggregate-target leakage.

Sources:

- Deccan Chronicle (2025): <https://www.deccanchronicle.com/nation/stubble-burning-cases-cross-500-mark-in-punjab-ppcb-data-1912152>
- India TV (Nov 2025): <https://www.indiatvnews.com/news/india/amid-poor-air-quality-in-delhi-punjab-sees-351-farm-fires-taking-total-count-to-3-284-since-september-sangrur-reports-highest-farm-fires-at-557-2025-11-07-1016227>
- Business Standard (Oct 2025): https://www.business-standard.com/india-news/stubble-burning-cases-in-punjab-cross-500-despite-govt-efforts-ppcb-data-125102301366_1.html
- ETV Bharat (Feb 2025) — confirms 2021 figure of 71,304 also matches Deccan Chronicle

All four sources independently report identical year-by-year totals: 50,590 / 55,210 / 76,590 / 71,304 / 49,922 / 36,663 / 10,909 (the last for 2024). This consistency across publications suggests the PPCB figures are authoritative.

Variables we did NOT include and why

- **District-level subsidy disbursements:** Punjab govt publishes district-totals only for some years. Incomplete data → would create biased features.
- **PUSA-44 cultivation restriction:** This was a 2024 policy. Since our test set ends in 2023, the dummy would be all zeros and provide no signal. Add when extending to 2024.

- **Variety mix (PR-126 vs PUSA-44 share):** We attempted to find year-by-year district-level variety adoption but published data is fragmented. Best available is CEEW's aggregate finding that short-duration variety acreage rose from 32.6% (2012) to 69.8% (2021). Could add as a single linear trend variable but signal is weak vs effort.
 - **Diesel prices:** Best added separately from PPAC weekly retail price. Not in this CSV.
 - **FIRMS confidence values for ground-truth weighting:** Already handled in upstream feature pipeline (v2 conf_weight column).
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Usage in modeling pipeline

1. Load: `policy = pd.read_csv('punjab_policy_2018_2023.csv')`
2. Drop the leakage column: `policy = policy.drop(columns=['ppcb_total_fires'])`
3. Merge: `df = df.merge(policy, on='year', how='left')`
4. Add to feature list. Most useful new features expected to be:
 - `crm_funds_cumulative_cr` (continuous, captures compounding mechanization)
 - `super_seeder_available` (binary, captures 2020 regime change)
 - `ngt_enforcement_level` (ordinal, captures regulatory pressure)
 - `msh_paddy_common` (continuous, captures cultivation incentive)
 - `ex_gratia_announced` (binary, captures 2019 specifically)
 - `election_year` (binary, captures 2022 specifically)

Constants across the period (`happy_seeder_subsidy_indiv_pct`, `happy_seeder_subsidy_coop_pct`) contribute zero variance and can be dropped before modeling. Keep them in the CSV for documentation and future extension.

Caveats for the deck

1. **Year-level granularity.** With 6 years, we have 6 unique policy environments. This limits our ability to causally identify policy effects. Frame these features as "controls" not "treatments."
2. **Some values are imputed.** Yearly splits of CRM funds are estimated from cumulative totals. Document this in the methodology slide.
3. **NGT enforcement is researcher-judgment.** Make this transparent. Alternative: use `ppcb_total_fires` (carefully, only for ranking enforcement intensity, never as a feature).
4. **Confounding.** 2022 is both the election year AND the year burning dropped sharply. Hard to separate election effect from underlying mechanization momentum. Acknowledge this.

5. **No causal claims.** Adding policy variables makes the model "policy-aware" but does not prove policy CAUSED any observed outcome. The deck should state this explicitly.